3-26E

3–26E Complete this table for refrigerant-134a:

T, °F	P, psia	h, Btu/lbm	x	Phase description
	80	78		
15			0.6	
10	70			
	180	129.46		
110			1.0	

T (°F)	P (psia)	h (Btu/lbm)	$x = \frac{h-h_f}{h_{fg}}$	Phase description
65.89	80	78	**0.46**	**saturated mixture**
15	**29.759**	**69.92**	0.6	**saturated mixture**
10	70	$\approx h_f =$ **15.308**	–	**compressed liquid,** $P > P_{\mathbf{sat}}$
117.69	180	129.46	–	**superheated vapor**
110	**161.16**	**117.25**	1.0	**saturated vapor**

For phase, compare the enthalpy to the saturated liquid and vapor enthalpies. If $h < h_f$, the phase is compressed liquid. If $h > h_{fg}$, the phase is superheated vapor. If $h_f < h < h_{fg}$, the phase is saturated mixture.

$$h_{\text{avg}} = h_f + x(h_{fg})$$

3-37

3–37 One kilogram of water vapor at 200 kPa fills the 1.1989-m^3 left chamber of a partitioned system shown in Fig. P3–37. The right chamber has twice the volume of the left and is initially evacuated. Determine the pressure of the water after the partition has been removed and enough heat has been transferred so that the temperature of the water is 3°C.

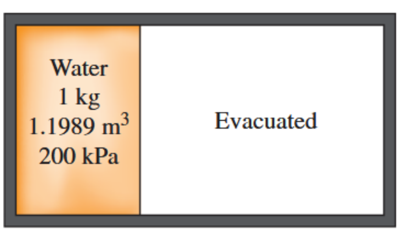

FIGURE P3–37

Given:

- mass $m = 1$ kg
- pressure $P = 200$ kPa
- volume $\cancel{V} = 1.1989$ m^3
- water vapor
- Right chamber has volume $2\cancel{V}$ and is evacuated

Find:

- Pressure of water after partition is open and it has temp $T = 3$°C

State 1:

$$\cancel{v}_1 = \frac{\cancel{V}}{m} = \frac{1.1989}{1} = 1.1989 \text{ m}^3/\text{kg}$$

State 2:

$$\nu_2 = \frac{3\mathcal{V}}{m} = \frac{3(1.1989)}{1} = 3.5967 \text{ m}^3/\text{kg}$$

Comparing the specific volume at state 2 to ν_f and ν_g, we see that it is a saturated mixture, so we can use table A-4 to find the saturation pressure (via linear interpolation):

$$P - P_0 = m(T - T_0) \quad \implies P - [0.8725\,\text{kPa}] = (0.0522)\left([3°C] - [5°C]\right)$$

$$\boxed{P_2 = P_{\text{sat @ }3°C} = 0.7679\,\text{kPa}}$$

3-43

3–43 100 kg of R-134a at 200 kPa are contained in a piston–cylinder device whose volume is 12.322 m^3. The piston is now moved until the volume is one-half its original size. This is done such that the pressure of the R-134a does not change. Determine the final temperature and the change in the total internal energy of the R-134a.

Given:

- State 1: m = 100 kg, P = 200 kPa, $\mathcal{V}_1$ = 12.322 m^3, R-134a
- State 2: m = 100 kg, P = 200 kPa, $\mathcal{V}_2$ = 0.5(12.322) m^3, R-134a

Find:

- Temperature at state 2 and change in internal energy from state 1 to state 2

At state 1,

$$\nu_1 = \frac{[12.322\,\text{m}^3]}{[100\,\text{kg}]} = 0.12322 \text{ m}^3/\text{kg}$$

Comparing to the saturated liquid and vapor specific volumes, we see that state 1 is a superheated vapor. Using table A-13, the point ν = 0.12322 m^3/kg at P=200 kPa = 0.2 MPa is given as $u_1 = 263.09\,\text{kJ/kg}$.

At state 2,

$$\nu_2 = \frac{[0.5(12.322)\,\text{m}^3]}{[100\,\text{kg}]} = 0.06161 \text{ m}^3/\text{kg}$$

Comparing to the saturated liquid and vapor specific volumes, we see that state 2 is a saturated mixture. Thus, its temperature is the saturation temperature at 200 kPa.

$$\boxed{T_2 = T_{\text{sat @200 kPa}} = -10.09°\text{C}}$$

To find internal energy at state 2, we can use the quality:

$$x = \frac{\nu_{\text{avg}} - \nu_f}{\nu_g - \nu_f} = \frac{[0.06161\,\text{m}^3/\text{kg}] - [0.0007532\,\text{m}^3/\text{kg}]}{[0.099951\,\text{m}^3/\text{kg}] - [0.0007532\,\text{m}^3/\text{kg}]} = 0.6134$$

Thus, the internal energy at state 2 is

$$u_2 = u_f + x(u_g - u_f) = 38.26 + 0.6134(186.25) = 152.505\,\mathrm{kJ/kg}$$

Finally, the change in internal energy from state 1 to state 2 is:

$$\Delta u = u_2 - u_1 = 152.505 - 263.09 = \boxed{-110.585\,\mathrm{kJ/kg}}$$

3-60

3–60 Water is being heated in a vertical piston–cylinder device. The piston has a mass of 40 kg and a cross-sectional area of 150 cm^2. If the local atmospheric pressure is 100 kPa, determine the temperature at which the water starts boiling.

Given:

- Water is heated under the weight of a 40 kg mass with area 150 cm^2 on top of it.
- $P_{\mathrm{atm}} = 100$ kPa

Find:

- Temperature at which water boils at the fixed pressure(saturation temperature)

The pressure exerted by the mass is given by

$$P_{\mathrm{mass}} = \frac{mg}{A} = \frac{[40\,\mathrm{kg}][9.81\,\mathrm{m/s}^2]}{[150\,\mathrm{cm}^2]\left[\frac{(10^{-2}\,\mathrm{m})^2}{1\,\mathrm{cm}^2}\right]} = 26160\,\mathrm{Pa} = 26.16\,\mathrm{kPa}$$

Thus, the total pressure on the water is

$$P = P_{\mathrm{atm}} + P_{\mathrm{mass}} = 100 + 26.16 = 126.16\,\mathrm{kPa}$$

At this pressure, the saturation temperature is found on table A-5 via linear interpolation (points: 125 kPa, 105.97°C, 150 kPa, 111.35°C):

$$T - T_0 = m(P - P_0) \quad \Longrightarrow \quad T - [105.97^\circ\mathrm{C}] = (0.2152)([126.16\,\mathrm{kPa}] - [125\,\mathrm{kPa}])$$

$$\boxed{T_{\mathrm{sat\ @\ 126.16\ kPa}} = 106.219^\circ\mathrm{C}}$$

3-69

3–69 A 100-L container is filled with 1 kg of air at a temperature of 27°C. What is the pressure in the container?

Given:

- $\mathcal{V} = 100$ L, $m = 1$ kg, $T = 27^\circ$C, air

Find:

- Pressure

Assuming that air acts as an ideal gas, we can find pressure using the ideal gas law:

$$\cancel{V} = [100\,\mathrm{L}]\left[\frac{1\,\mathrm{m}^3}{1000\,\mathrm{L}}\right] = 0.1\,\mathrm{m}^3$$

$$T = [27°\mathrm{C}] + 273.15 = 300.15\,\mathrm{K}$$

$$P\cancel{V} = mRT \quad \Longrightarrow \quad P = \frac{mRT}{\cancel{V}} = \frac{[1\,\mathrm{kg}][0.2870\,\mathrm{kJ/kg\cdot K}][300.15\,\mathrm{K}]}{[0.1\,\mathrm{m}^3]} = \boxed{861.0\,\mathrm{kPa}}$$

3-75

3–75 A 1-m^3 tank containing air at 10°C and 350 kPa is connected through a valve to another tank containing 3 kg of air at 35°C and 150 kPa. Now the valve is opened, and the entire system is allowed to reach thermal equilibrium with the surroundings, which are at 20°C. Determine the volume of the second tank and the final equilibrium pressure of air. *Answers:* 1.77 m^3, 222 kPa

Given:

- State 1:
  - A: $\cancel{V}_{1A} = 1$ m^3, $T_{1A} =$ 10°C, $P_{1A} = 350$ kPa
  - B: $m_{1B} = 3$ kg, $T_{1B} =$ 35°C, $P_{1B} = 150$ kPa
- State 2: A and B mix and reach thermal equilibrium at $T_2 =$ 20°C

Find:

- Volume of B at state 1, $\cancel{V}_{1B}$, and final pressure P_2 at state 2

At state 1, we can find the volume of B using the ideal gas law:

$$T_{1B} = [35°\mathrm{C}] + 273.15 = 308.15\,\mathrm{K}$$

$$P\cancel{V} = mRT \quad \Longrightarrow \quad \cancel{V}_{1B} = \frac{m_{1B}RT_{1B}}{P_{1B}} = \frac{[3\,\mathrm{kg}][0.2870\,\mathrm{kJ/kg\cdot K}][308.15\,\mathrm{K}]}{[150\,\mathrm{kPa}]} = \boxed{1.768\,\mathrm{m}^3}$$

Finding mass of A using ideal gas law:

$$T_{1A} = [10°\mathrm{C}] + 273.15 = 283.15\,\mathrm{K}$$

$$P\cancel{V} = mRT \quad \Longrightarrow \quad m_{1A} = \frac{P_{1A}\cancel{V}_{1A}}{RT_{1A}} = \frac{[350\,\mathrm{kPa}][1\,\mathrm{m}^3]}{[0.2870\,\mathrm{kJ/kg\cdot K}][283.15\,\mathrm{K}]} = 4.306\,\mathrm{kg}$$

At state 2, the masses and volumes of A and B combine (it stays constant), using the given temperature, we can use the ideal gas law to find the final pressure:

$$T_2 = [20°\mathrm{C}] + 273.15 = 293.15\,\mathrm{K}$$

$$\cancel{V}_2 = \cancel{V}_{1A} + \cancel{V}_{1B} = 1 + 1.768 = 2.768\,\mathrm{m}^3$$

$$P_2 = \frac{m_2RT_2}{\cancel{V}_2} = \frac{[4.306 + 3\,\mathrm{kg}][0.2870\,\mathrm{kJ/kg\cdot K}][293.15\,\mathrm{K}]}{[2.768\,\mathrm{m}^3]} = \boxed{222.096\,\mathrm{kPa}}$$

3-95E

3–95E Refrigerant-134a at 400 psia has a specific volume of 0.1144 ft^3/lbm. Determine the temperature of the refrigerant based on (*a*) the ideal-gas equation, (*b*) the van der Waals equation, and (*c*) the refrigerant tables.

(a)

$$R = 0.08149 \left[\frac{\mathrm{kJ}}{\mathrm{kg}\cdot\mathrm{K}}\right]\left[\frac{778.169\,\mathrm{ft}\cdot\mathrm{lbf}}{1055.06\,\mathrm{J}}\right]\left[\frac{1\,\mathrm{J}}{1\,\mathrm{kJ}}\right][]$$

3-136

3–136 A 3-m^3 rigid vessel contains steam at 2 MPa and 500°C. The mass of the steam is

(*a*) 13 kg (*b*) 17 kg (*c*) 22 kg
(*d*) 28 kg (*e*) 35 kg

Given:

-

Find:

-